RetailPulse — MySQL Data Analysis SQL Questions

Database: `retailpulse_dw` • Tables: `regions, customers, categories, suppliers, products, campaigns, orders, order_items, returns`

Part 1: Data Cleaning & Validation Questions (6 Queries)

A1. Write a single SQL query to display the row count of all 9 tables (`regions` through `returns`).

A2. Audit `customers` for missing names, invalid email formats, missing birth dates, and missing signup dates; audit `orders` for delivered orders without `delivered_date` , cancelled orders with a non-null `shipped_date` , and negative `order_discount` or `shipping_fee` .

A3. Detect any duplicate customer emails, duplicate product SKUs, or orders containing the same `product_id` across multiple `order_items` rows.

A4. Find chronological and financial rule violations: (a) orders placed before a customer's `signup_date` , (b) orphan orders with no line items, and (c) returns where `return_date < DATE(delivered_date)` , `refund_amount` exceeds line net amount, or `order_id` mismatches.

A5. Identify any pricing or margin anomalies where billed `order_items.unit_price` differs from catalog `products.unit_price` or where `unit_price < unit_cost` .

A6. Build a consolidated Data Health Summary table showing `audit_check` , `issue_count` , and `status` (`PASS` or `REVIEW`) covering signup chronology, orphan orders, return date/refund validity, negative margins, and order-level over-refunds after `order_discount` .

Part 2: Exploratory & Core KPI SQL Questions (Q1–Q12)

Q1. Compute the Executive Summary KPI Scorecard: Total Orders, Active Customers, Delivered Orders, Cancelled Orders, Fully Returned Orders, Cancellation Rate %, Gross Merchandise Value (GMV), Total Refunds, Realized Net Revenue, Total COGS, Gross Profit, Gross Margin %, and Average Order Value (AOV).

Q2. Calculate Monthly Revenue, Order Count, Gross Profit, Prior Month Revenue, and Month-over-Month (MoM) Revenue Growth % ordered by month (excluding cancelled orders).

Q3. Perform a Year-over-Year (YoY) Quarterly comparison showing Order Year, Quarter (`Q1` – `Q4`), Total Orders, Quarterly Revenue, Quarterly Profit, Margin %, Prior Year Same-Quarter Revenue, and YoY Revenue Growth %.

Q4. Analyze Category Performance: Units Sold, Orders with Category, Gross Revenue, Refunds, Net Revenue, Gross Profit, Actual Margin %, Target Margin %, Margin Variance (pts), and Net Revenue Share %.

Q5. Evaluate Regional Performance & Delivery SLAs: Total Orders, Non-Cancelled Orders, Net Revenue, Avg Days to Ship, Avg Days to Deliver, Late Deliveries Count (exceeding `promised_days`), and Late Delivery %.

Q6. Analyze Payment Methods: Total Orders, Order Share %, Net Revenue, Avg Order Value (AOV), Cancellation Rate %, and Full-Order Return Rate %.

Q7. Find the Top 10 Products by Net Revenue along with Units Sold, Net Revenue, Gross Profit, Profit Margin %, overall Revenue Rank, and overall Profit Rank.

Q8. Build a Supplier Scorecard showing Supplier Name, Country, Lead Time Days, Reliability Score, Active SKUs, Units Sold, Gross Sales, Returned Line Items, and Item Return Rate %.

Q9. Compare `Discounted` orders vs. `Full Price` orders across Order Count, Avg Gross Basket, Avg Discount Given, Avg Net Order Value, Total Net Revenue, Total Gross Profit, and Gross Margin %.

Q10. Calculate Marketing Campaign Performance for all 8 campaigns: Campaign Spend, Attributed Orders, Unique Buyers, Attributed Realized Net Revenue, Gross Profit Generated, ROAS (x), and Net Marketing ROI %.

Q11. Break down Returns by `return_reason`: Return Count, Share of Returns %, Total Refund Leakage, Restocked Items Count, and Damaged Write-Offs Count.

Q12. Analyze Day-of-Week ordering patterns (Sunday through Saturday in calendar order) showing Total Orders, Total Revenue, and Average Order Value.

Part 3: Advanced Analytics — RFM, Cohorts & Market Basket (Q13–Q24)

Q13. Create a Top 15 Customer Lifetime Value (CLV) Leaderboard displaying Customer Name, Region, Loyalty Tier, Total Orders, First & Last Order Dates, Realized LTV, AOV, Global LTV Rank, and Regional LTV Rank.

Q14. Build a full RFM Segmentation model using `NTILE(5)` (relative to the max non-cancelled order date): compute Recency Days, Frequency, Monetary Value, R/F/M scores (1–5), 3-digit RFM Code, Avg RFM Score, and classify into `Champions`, `Loyal Customers`, `Recent / Potential Loyalists`, `At Risk`, `About to Sleep`, `Hibernating / Lost`, or `Needs Attention`.

Q15. Summarize the RFM Segments showing Customer Count, Customer Share %, Avg Recency Days, Avg Frequency, Avg LTV, Segment Total Revenue, and Revenue Share %.

Q16. Build a Quarterly Cohort Retention Matrix grouping buyers by their first non-cancelled order quarter and reporting retention % for `Q+1`, `Q+2`, `Q+3`, `Q+4`, and `Q+5+`.

Q17. Perform Repeat Purchase Gap Analysis for multi-order customers using `LAG()` to find Repeat Order Transitions, Min Gap Days, Avg Gap Days, and Max Gap Days between consecutive orders.

Q18. Detect Personalized Churn Risk for repeat buyers (2+ orders) by comparing `days_inactive` against their personal `avg_cycle_days` to compute `overdue_ratio` and flag `High Churn Risk` (>2.0x), `Watchlist` (>1.25x), or `Healthy Cadence`.

Q19. Perform ABC Pareto Product Classification using cumulative net revenue % ordered descending: `Class A (Core Stars)` (≤70%), `Class B (Moderate)` (≤90%), and `Class C (Long Tail)` (>90%).

Q20. Perform Market Basket Analysis to find the Top 12 distinct product pairs bought together in at least 2 non-cancelled orders, calculating Co-Purchase Count, Support %, Confidence A→B %, Confidence B→A %, and Lift Score.

Q21. Compute Monthly Revenue alongside Cumulative Running Revenue and a 3-Month Rolling Moving Average Revenue.

Q22. Evaluate Acquisition Channel Quality across all 6 channels: Total Customers, Repeat Buyers (2+ orders), Repeat Buyer Rate %, Avg Orders per Customer, Avg Customer LTV, and Total Channel LTV.

Q23. Build an Inventory Stock-Out Risk & Reorder Alert report using trailing 180-day sales velocity, Daily Run Rate, and Estimated Days of Cover to flag `REORDER NOW`, `LOW BUFFER`, or `HEALTHY`.

Q24. Build a Customer Order Conversion Funnel across all signed-up customers: Total Signups, ≥1 Order, ≥2 Orders, ≥3+ Orders, and stage-to-stage conversion %.

Part 4: Views, Stored Function, Stored Procedures & Trigger (B1–B8)

B1. Write a deterministic Stored Function `fn_profit_margin_pct(p_revenue, p_cost)` that safely returns profit margin % (or `0.00` when revenue ≤ 0).

B2. Create View `vw_order_financials` returning order-level GMV, total discounts, refunds, realized net revenue, COGS, gross profit, margin %, actual delivery days, and SLA status.

B3. Create View `vw_customer_360` summarizing every customer's completed/cancelled orders, first/last order date, recency, lifetime net revenue, profit, refunds, AOV, R/F/M scores, and RFM segment.

B4. Create View `vw_product_performance` summarizing SKU units sold, gross revenue, refunds, net revenue, COGS, returned line count, and item return rate %.

B5. Create Stored Procedure `sp_get_executive_kpis(p_start_date, p_end_date)` returning KPI metrics for any custom date range.

B6. Create Stored Procedure `sp_get_category_drilldown(p_category_name, p_year)` returning monthly SKU performance for a category (supporting `NULL` for all years).

B7. Create Stored Procedure `sp_get_customers_by_rfm_segment(p_segment_name)` returning customers in a selected RFM segment ordered by LTV.

B8. Create Trigger `trg_after_order_item_insert` that safely decrements `products.stock_qty` and inserts an entry into `inventory_audit_log` whenever a non-cancelled order item is inserted.